

Model Refinement and Test Submission

Frozen, calibrated logistic churn-proxy candidate and untouched temporal evaluation

Course: Samsung Innovation Campus — AI in Marketing Capstone

Team: Group 7

Project: GrowthPilot AI

Prepared by: Şahin Başcı

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AI AUTHORSHIP DISCLOSURE — This document was drafted with OpenAI Codex/ChatGPT assistance and checked against repository evidence. Reported project metrics come from executed, versioned artifacts; external claims are cited. Human review remains required.

Part I — Model refinement

1. Initial evaluation and weaknesses

Initial exploration selected logistic regression by validation PR-AUC 0.805322. Its weaknesses were probability miscalibration (Brier 0.213300; ECE 0.169377) and sensitivity to regularization. LightGBM was nearly tied on PR-AUC and had better raw calibration, so refinement compared all three learned families without changing the primary selection rule.

2. Refinement techniques

- Logistic grid: $C \in \{0.01, 0.05, 0.2, 1.0, 5.0\}$; class weighting kept off because ranking/calibration and capacity metrics were evaluated directly.
- Random forest grid: 400 trees, minimum leaf {5,10,20}, maximum depth {8,16}.
- LightGBM: 12 seeded search trials across learning rate, leaves, depth, minimum child samples, L2 regularization, row and column subsampling.
- Sigmoid calibration fitted only on the 2011-06-01 calibration snapshot.
- Operating threshold fixed at 0.812509305136 from approximately 10% calibration capacity.

3. Tuning impact

Stage	PR-AUC	ROC-AUC	Brier	ECE	Top-10% precision / lift
Initial logistic validation	0.805322	0.782819	0.213300	0.169377	0.871642 / 1.510154

Stage	PR-AUC	ROC-AUC	Brier	ECE	Top-10% precision / lift
Refined logistic C=.01 validation	0.812223	0.787127	0.196710	0.106133	0.883582 / 1.530841
Calibration-fit descriptive	0.739014	0.773722	0.193632	0.040863	0.834586 / 1.671058

Calibration-fit metrics reuse calibration labels and are descriptive, not unbiased test estimates.

Lowering C from 1.0 to 0.01 increased validation PR-AUC to 0.812223 and reduced Brier to 0.196710.

Sigmoid calibration on the later calibration split reduced ECE to 0.040863 descriptively. The final serialized candidate checksum is

942d705d66a9469d57494626a99da83c91a9a895a3a2b2d7e977ff3ba56952ad.

4. Cross-validation and feature selection decisions

Random k-fold cross-validation was not used because it would mix future and past snapshots and repeated customers. The design instead uses expanding chronological training cutoffs plus distinct validation, calibration and final-test dates. No post-hoc feature selection was performed: the regularized logistic pipeline controls coefficient magnitude, and removing features after observing the test would violate the freeze.

5. Explainability validation

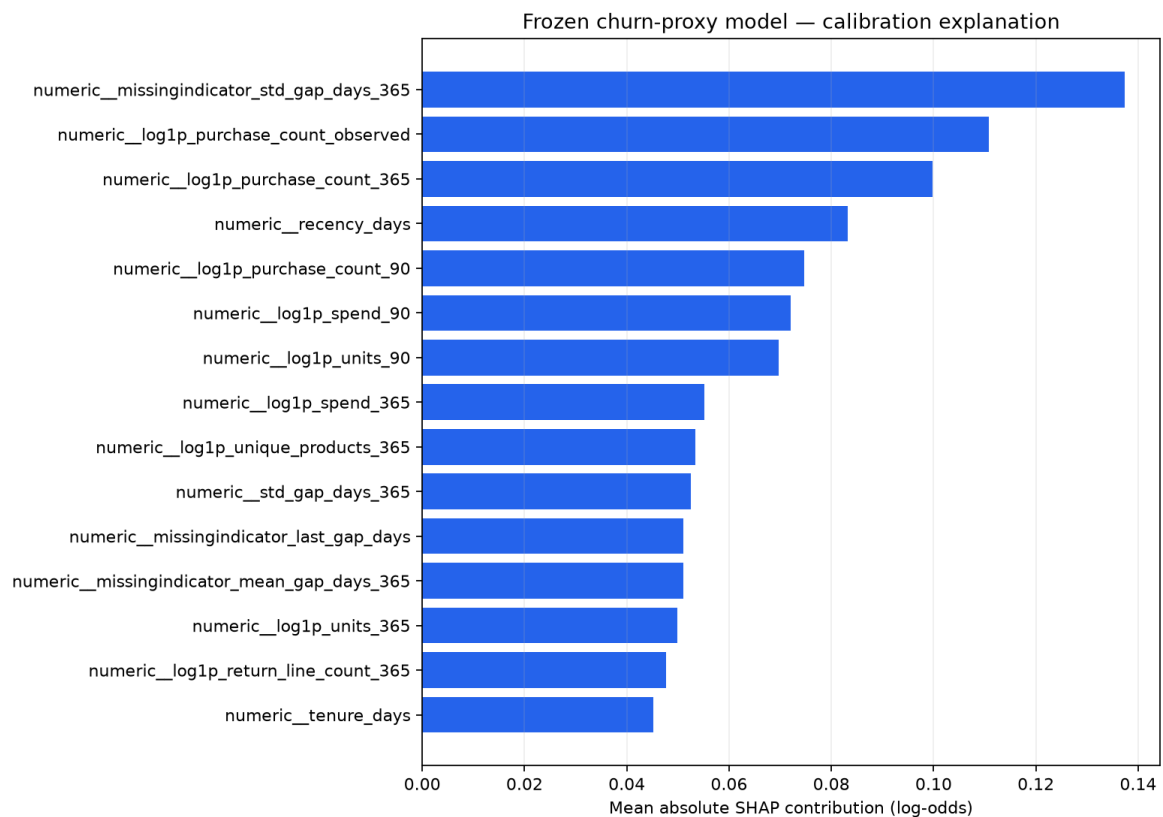


Figure 1. Global mean absolute SHAP contribution in calibrated log-odds. Project-generated diagnostic.

The SHAP decomposition was computed in calibrated log-odds for the frozen logistic pipeline. Maximum additivity error is $8.881784197001252 \times 10^{-16}$, confirming numerical consistency. Contributions explain this model's score formation; they are not causal effects and must not be turned into unsupported personal claims.

Part II — Final test submission

1. Test preparation and integrity

The final test is the 2,772-customer snapshot at cutoff 2011-09-01 with a fully observed 90-day label window. Its prepared-file SHA-256 is 2830e4738acfee8c5561f5eb513874d0cacae32f3b3095c412db62dc2cf78ff6. Before access, the candidate manifest, preprocessing, sigmoid calibrator, artifact checksum and threshold were frozen. Post-test retuning is prohibited.

2. Model application

```
bundle = joblib.load('artifacts/models/final_candidate.joblib')
assert sha256_file(model_path) == EXPECTED_MODEL_SHA256
scores = bundle['calibrated_model'].predict_proba(X_test)[: , 1]
predicted = scores >= 0.812509305136
# target, feature and split versions are checked before reporting
```

3. Final test metrics

Measure	Final test result	95% bootstrap interval
Rows / prevalence	2,772 / 0.392857	Not applicable
PR-AUC	0.647525	[0.618131, 0.678937]
ROC-AUC	0.765878	[0.748534, 0.782761]
Brier score	0.199854	[0.192598, 0.207125]
Log loss	0.582761	Not bootstrapped
ECE (10 bins)	0.095054	Not bootstrapped
Top-10% precision / recall / lift	0.748201 / 0.191001 / 1.904513	Not bootstrapped

Evidence class: project-executed, untouched temporal holdout. Intervals are nonparametric row bootstrap intervals for this cutoff.

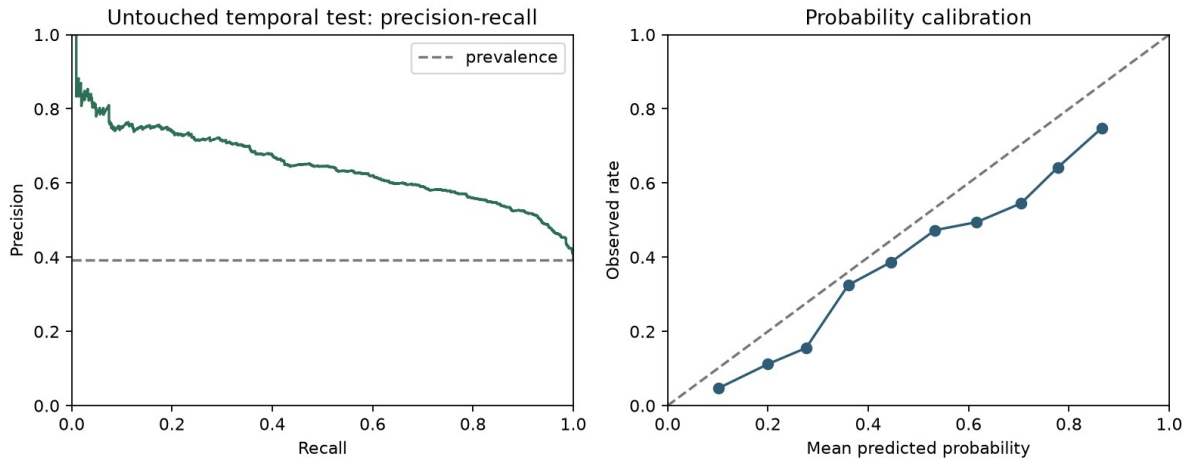


Figure 2. Final temporal-test precision-recall and probability calibration.

4. Frozen-threshold confusion matrix

Actual / predicted	Predicted active	Predicted inactive
Actual active	TN = 1,603	FP = 80
Actual inactive	FN = 867	TP = 222

Threshold 0.812509305136 selected 302 customers (10.8947%); precision 0.735099, recall 0.203857, F1 0.319195.

5. Validation-to-test comparison

Split	PR-AUC	ROC-AUC	Brier	Prevalence	Interpretation
Validation	0.812223	0.787127	0.196710	0.577187	Used for family/tuning selection
Calibration	0.739014	0.773722	0.193632	0.499436	Used for sigmoid and threshold; descriptive
Final test	0.647525	0.765878	0.199854	0.392857	Single untouched temporal holdout

PR-AUC falls from validation 0.812223 to test 0.647525 while prevalence also falls from 0.577187 to 0.392857. ROC-AUC remains 0.765878 and top-10% lift rises to 1.904513. The gap is evidence of temporal/data shift and uncertainty, not a reason to revise the frozen test result.

6. Deployment truth

The artifact is registered locally as growthpilot-churn-inactivity version 1 with alias champion and is integrated into tenant-scoped on-demand and durable batch-scoring services. “Champion” is a local registry alias, not proof of production deployment. On-demand scoring is demo-gated; live cloud, tenant-data drift and service-level performance are unvalidated. Every score remains non-authorizing.

7. Conclusion

The frozen model provides useful ranking signal on the historical temporal test, especially within limited outreach capacity, but misses most inactive customers at the strict threshold. It should support review,

not replace human judgment. Production adoption requires current tenant data, lawful use, monitored calibration/drift and a causal campaign evaluation.

References and evidence

scikit-learn developers. Probability calibration and model evaluation documentation. <https://scikit-learn.org/stable/modules/calibration.html>

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. NeurIPS 30. <https://arxiv.org/abs/1705.07874>

Project evidence: artifacts/ml/final_candidate.json, final_evaluation.json/png, explanations.json/png, model_registry.json; scripts/refine_model.py, evaluate_final.py, explain_model.py and register_model.py.